

## WINDER BORROW, GA, USA

WMO: 747809

Lat: 33.983N

Lon: 83.668W

Elev: 287

StdP: 97.92

Time Zone: -5.00 (NAE)

Period: 02-19

WBAN: 03832

## Annual Heating, Humidification, and Ventilation Design Conditions

| Coldest Month | Heating DB |      | Humidification DP/MCDB and HR |     |      |       |     |      | Coldest Month WS/MCDB |      |     |      | MCWS/PCWD to 99.6% DB |      | WSF   |
|---------------|------------|------|-------------------------------|-----|------|-------|-----|------|-----------------------|------|-----|------|-----------------------|------|-------|
|               |            |      | 99.6%                         |     |      | 99%   |     |      | 0.4%                  |      | 1%  |      |                       |      |       |
|               | 99.6%      | 99%  | DP                            | HR  | MCDB | DP    | HR  | MCDB | WS                    | MCDB | WS  | MCDB | MCWS                  | PCWD |       |
| (a)           | (b)        | (c)  | (d)                           | (e) | (f)  | (g)   | (h) | (i)  | (j)                   | (k)  | (l) | (m)  | (n)                   | (o)  | (p)   |
| 1             | -4.1       | -2.2 | -17.3                         | 0.8 | 3.3  | -13.0 | 1.3 | 2.0  | 10.3                  | 4.5  | 9.0 | 4.6  | 2.6                   | 320  | 0.360 |

## Annual Cooling, Dehumidification, and Enthalpy Design Conditions

| Hottest Month | Hottest Month DB Range | Cooling DB/MCWB |      |      |      |      |      | Evaporation WB/MCDB |      |      |      |      |      | MCWS/PCWD to 0.4% DB |      |
|---------------|------------------------|-----------------|------|------|------|------|------|---------------------|------|------|------|------|------|----------------------|------|
|               |                        | 0.4%            |      | 1%   |      | 2%   |      | 0.4%                |      | 1%   |      | 2%   |      |                      |      |
|               |                        | DB              | MCWB | DB   | MCWB | DB   | MCWB | WB                  | MCDB | WB   | MCDB | WB   | MCDB | MCWS                 | PCWD |
| (a)           | (b)                    | (c)             | (d)  | (e)  | (f)  | (g)  | (h)  | (i)                 | (j)  | (k)  | (l)  | (m)  | (n)  | (o)                  | (p)  |
| 7             | 10.2                   | 35.0            | 22.7 | 33.7 | 22.8 | 32.5 | 22.7 | 25.7                | 31.2 | 25.2 | 30.8 | 24.7 | 30.0 | 2.5                  | 290  |

| Dehumidification DP/MCDB and HR |      |      |      |      |      |      |      |      | Enthalpy/MCDB |      |      |      |      |      | Extreme<br>Max WB |
|---------------------------------|------|------|------|------|------|------|------|------|---------------|------|------|------|------|------|-------------------|
| 0.4%                            |      |      | 1%   |      |      | 2%   |      |      | 0.4%          |      | 1%   |      | 2%   |      |                   |
| DP                              | HR   | MCDB | DP   | HR   | MCDB | DP   | HR   | MCDB | Enth          | MCDB | Enth | MCDB | Enth | MCDB |                   |
| (a)                             | (b)  | (c)  | (d)  | (e)  | (f)  | (g)  | (h)  | (i)  | (j)           | (k)  | (l)  | (m)  | (n)  | (o)  | (p)               |
| 24.1                            | 19.7 | 27.6 | 23.3 | 18.7 | 27.0 | 22.9 | 18.2 | 26.7 | 80.5          | 31.4 | 78.4 | 30.4 | 76.1 | 30.4 | 32.0              |

## Extreme Annual Design Conditions

| Extreme Annual WS |      |     | Extreme Annual Temperature |      |                    |     | n-Year Return Period Values of Extreme Temperature |      |            |      |            |      |            |      |    |    |
|-------------------|------|-----|----------------------------|------|--------------------|-----|----------------------------------------------------|------|------------|------|------------|------|------------|------|----|----|
|                   |      |     | Mean                       |      | Standard Deviation |     | n=5 years                                          |      | n=10 years |      | n=20 years |      | n=50 years |      |    |    |
| 1%                | 2.5% | 5%  | Min                        | Max  | Min                | Max | Min                                                | Max  | Min        | Max  | Min        | Max  | Min        | Max  | DB | WB |
| (a)               | (b)  | (c) | (e)                        | (f)  | (g)                | (h) | (i)                                                | (j)  | (k)        | (l)  | (m)        | (n)  | (o)        | (p)  |    |    |
| (4) 8.1           | 7.0  | 5.6 | -7.7                       | 36.7 | 1.6                | 2.0 | -8.9                                               | 38.1 | -9.8       | 39.2 | -10.7      | 40.3 | -11.9      | 41.8 |    |    |
| (5)               |      |     | -9.0                       | 26.8 | 1.5                | 1.8 | -10.0                                              | 28.0 | -10.9      | 29.1 | -11.7      | 30.1 | -12.8      | 31.3 |    |    |

## Monthly Climatic Design Conditions

|                                                                        |             | Annual | Jan   | Feb   | Mar   | Apr   | May   | Jun   | Jul   | Aug   | Sep   | Oct   | Nov   | Dec   |  |  |
|------------------------------------------------------------------------|-------------|--------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|--|--|
|                                                                        |             | (d)    | (e)   | (f)   | (g)   | (h)   | (i)   | (j)   | (k)   | (l)   | (m)   | (n)   | (o)   | (p)   |  |  |
| (6) Temperatures, Degree-Days and Degree-Hours                         | DBAvg       | 17.2   | 6.5   | 8.3   | 12.8  | 16.8  | 21.2  | 25.4  | 26.8  | 26.4  | 23.5  | 17.6  | 12.0  | 8.4   |  |  |
|                                                                        | DBStd       | 8.12   | 5.08  | 4.63  | 4.67  | 3.95  | 3.46  | 2.44  | 2.15  | 2.02  | 2.93  | 4.15  | 4.26  | 4.46  |  |  |
|                                                                        | HDD10.0     | 357    | 132   | 81    | 26    | 3     | 0     | 0     | 0     | 0     | 0     | 3     | 26    | 87    |  |  |
|                                                                        | HDD18.3     | 1484   | 366   | 283   | 179   | 73    | 16    | 1     | 0     | 0     | 2     | 63    | 193   | 309   |  |  |
|                                                                        | CDD10.0     | 2984   | 24    | 33    | 112   | 208   | 348   | 463   | 520   | 508   | 406   | 237   | 87    | 37    |  |  |
|                                                                        | CDD18.3     | 1071   | 0     | 1     | 8     | 28    | 106   | 213   | 262   | 250   | 158   | 39    | 5     | 1     |  |  |
|                                                                        | CDH23.3     | 9760   | 0     | 6     | 74    | 294   | 919   | 2024  | 2577  | 2301  | 1259  | 281   | 24    | 1     |  |  |
|                                                                        | CDH26.7     | 3936   | 0     | 0     | 7     | 53    | 293   | 872   | 1176  | 1007  | 469   | 58    | 1     | 0     |  |  |
| (14) Wind                                                              | WSAvg       | 2.1    | 2.7   | 2.7   | 2.7   | 2.4   | 2.0   | 1.7   | 1.6   | 1.3   | 1.6   | 1.9   | 2.1   | 2.4   |  |  |
| (15) Precipitation                                                     | PrecAvg     | 1324   | 124   | 114   | 145   | 90    | 95    | 122   | 122   | 109   | 108   | 79    | 103   | 97    |  |  |
|                                                                        | PrecMax     | 1623   | 212   | 214   | 241   | 238   | 230   | 260   | 287   | 221   | 310   | 214   | 210   | 210   |  |  |
|                                                                        | PrecMin     | 1016   | 64    | 40    | 30    | 40    | 35    | 34    | 36    | 36    | 0     | 8     | 26    | 54    |  |  |
|                                                                        | PrecStd     | 162    | 38    | 45    | 56    | 49    | 43    | 58    | 68    | 54    | 76    | 56    | 51    | 34    |  |  |
| (19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures | 0.4%        | DB     | 21.2  | 23.8  | 27.6  | 30.1  | 32.8  | 36.0  | 36.2  | 36.3  | 34.9  | 32.0  | 26.2  | 22.6  |  |  |
|                                                                        |             | MCWB   | 15.6  | 17.1  | 15.6  | 18.4  | 19.5  | 22.7  | 23.0  | 22.7  | 21.2  | 21.8  | 18.0  | 17.0  |  |  |
|                                                                        | 2%          | DB     | 18.0  | 20.9  | 25.9  | 27.9  | 31.3  | 34.0  | 35.0  | 34.8  | 32.7  | 27.8  | 22.9  | 20.3  |  |  |
|                                                                        |             | MCWB   | 14.8  | 14.6  | 15.6  | 17.2  | 20.3  | 22.5  | 23.0  | 22.8  | 21.2  | 19.6  | 16.5  | 16.7  |  |  |
|                                                                        | 5%          | DB     | 17.1  | 17.9  | 22.9  | 27.0  | 30.0  | 32.6  | 33.2  | 32.9  | 31.3  | 26.3  | 21.3  | 17.8  |  |  |
|                                                                        |             | MCWB   | 13.7  | 13.4  | 14.8  | 16.8  | 19.9  | 22.2  | 23.5  | 23.2  | 21.6  | 18.5  | 15.8  | 14.5  |  |  |
|                                                                        | 10%         | DB     | 14.8  | 16.3  | 21.2  | 25.0  | 27.8  | 31.3  | 32.4  | 32.0  | 29.9  | 24.8  | 19.8  | 16.2  |  |  |
|                                                                        |             | MCWB   | 11.6  | 11.3  | 14.0  | 16.0  | 19.6  | 22.3  | 23.5  | 23.1  | 21.0  | 18.2  | 14.7  | 13.3  |  |  |
| (27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures | 0.4%        | WB     | 17.9  | 18.9  | 19.2  | 21.1  | 23.9  | 26.3  | 26.6  | 26.3  | 25.2  | 23.9  | 21.7  | 19.5  |  |  |
|                                                                        |             | MCDB   | 19.2  | 21.8  | 22.8  | 24.9  | 28.6  | 31.9  | 32.1  | 31.4  | 28.8  | 27.8  | 23.3  | 21.3  |  |  |
|                                                                        | 2%          | WB     | 16.3  | 17.0  | 17.8  | 19.8  | 22.9  | 25.2  | 25.7  | 25.6  | 24.1  | 22.7  | 19.4  | 17.4  |  |  |
|                                                                        |             | MCDB   | 17.5  | 19.2  | 21.3  | 24.3  | 27.7  | 31.1  | 31.3  | 31.2  | 28.3  | 26.0  | 21.6  | 18.9  |  |  |
|                                                                        | 5%          | WB     | 14.5  | 15.0  | 16.6  | 18.8  | 22.1  | 24.5  | 25.1  | 25.1  | 23.4  | 21.5  | 17.4  | 16.0  |  |  |
|                                                                        |             | MCDB   | 16.2  | 17.0  | 20.2  | 23.7  | 26.8  | 29.8  | 30.7  | 30.7  | 27.6  | 24.1  | 19.3  | 17.7  |  |  |
|                                                                        | 10%         | WB     | 12.3  | 12.8  | 15.5  | 17.6  | 21.2  | 23.8  | 24.7  | 24.6  | 22.8  | 20.0  | 15.9  | 13.7  |  |  |
|                                                                        |             | MCDB   | 13.9  | 15.5  | 19.6  | 22.1  | 25.4  | 28.7  | 30.0  | 29.8  | 26.9  | 22.6  | 18.3  | 16.0  |  |  |
| (35) Mean Daily Temperature Range                                      | 5% DB       | MDBR   | 9.9   | 10.3  | 11.6  | 12.2  | 11.3  | 10.6  | 10.2  | 9.9   | 10.3  | 11.0  | 11.5  | 9.8   |  |  |
|                                                                        |             | MCDBR  | 11.8  | 12.4  | 13.8  | 14.5  | 13.0  | 12.2  | 11.4  | 11.8  | 12.3  | 12.7  | 13.1  | 11.5  |  |  |
|                                                                        |             | MCWBR  | 8.4   | 7.9   | 7.2   | 6.3   | 4.8   | 4.2   | 4.1   | 4.0   | 4.4   | 5.6   | 7.4   | 8.1   |  |  |
|                                                                        | 5% WB       | MCDBR  | 9.6   | 9.4   | 11.2  | 11.7  | 10.3  | 10.2  | 10.2  | 10.0  | 9.1   | 9.2   | 10.1  | 9.8   |  |  |
|                                                                        |             | MCWBR  | 8.3   | 7.7   | 7.4   | 6.3   | 4.9   | 4.1   | 4.0   | 3.6   | 4.3   | 5.3   | 7.5   | 8.5   |  |  |
| (40) Clear-Sky Solar Irradiance                                        | taub        |        | 0.311 | 0.316 | 0.351 | 0.388 | 0.447 | 0.476 | 0.516 | 0.514 | 0.418 | 0.376 | 0.339 | 0.309 |  |  |
|                                                                        | taud        |        | 2.524 | 2.503 | 2.432 | 2.315 | 2.189 | 2.151 | 2.059 | 2.050 | 2.303 | 2.407 | 2.460 | 2.550 |  |  |
|                                                                        | Ebn at Noon |        | 905   | 937   | 924   | 899   | 846   | 817   | 782   | 776   | 843   | 855   | 860   | 886   |  |  |
|                                                                        | Edh at Noon |        | 83    | 94    | 109   | 128   | 147   | 153   | 166   | 165   | 123   | 102   | 88    | 77    |  |  |
| (44) All-Sky Solar Radiation                                           | RadAvg      |        | 2.71  | 3.28  | 4.36  | 5.46  | 6.04  | 6.24  | 5.93  | 5.46  | 4.74  | 3.99  | 3.00  | 2.30  |  |  |
|                                                                        | RadStd      |        | 0.16  | 0.41  | 0.32  | 0.36  | 0.48  | 0.48  | 0.41  | 0.29  | 0.46  | 0.47  | 0.30  | 0.24  |  |  |

## Historical Trends

|                              | DBAvg | Heating |        | Cooling |       |       | Degree-Days |         |         |         |
|------------------------------|-------|---------|--------|---------|-------|-------|-------------|---------|---------|---------|
|                              |       | 99% DB  | 99% DP | 1% DB   | 1% WB | 1% DP | HDD10.0     | HDD18.3 | CDD10.0 | CDD18.3 |
|                              |       | (d)     | (e)    | (f)     | (g)   | (h)   | (i)         | (j)     | (k)     | (l)     |
| (46) Station Only            |       | N/A     | N/A    | N/A     | N/A   | N/A   | N/A         | N/A     | N/A     | N/A     |
| (47) Regional (41 neighbors) |       | +0.54   | N/A    | N/A     | N/A   | N/A   | N/A         | N/A     | +167    | +111    |

Nomenclature: See separate page